$$C_{Hl}^{(3)\,i1\_i2\_} \to \frac{1}{16\pi^2}\left(-\frac{1}{144}\frac{1}{m_\Phi^2}\, s_\gamma^2\, y_e^{i1p}\left(g_2^2\,\overline{y_e}^{i2p} + 12\, c_\gamma^2\,\overline{y_e}^{rp}\,\overline{y_e}^{i2s}\, y_e^{rs}\right) +\right.$$

$$\frac{1}{9} g_2^4\, \mathrm{LF}_{3,0}[m_2]\,\delta_{i1i2} + \frac{1}{6} g_2^4\, \mathrm{LF}_{4,-1}[m_2]\,\delta_{i1i2} - \frac{8}{45} g_2^4\, \mathrm{LF}_{5,-2}[m_2]\,\delta_{i1i2} +$$

$$\frac{1}{18}\sum_p g_2^4\, \mathrm{LF}_{3,0}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2} - \frac{5}{48}\sum_p g_2^4\, \mathrm{LF}_{4,-1}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2} + \frac{2}{45}\sum_p g_2^4\, \mathrm{LF}_{5,-2}\left[m_{\tilde{l}}^{p}\right]\delta_{i1i2} +$$

$$\frac{1}{6}\sum_p g_2^4\, \mathrm{LF}_{3,0}\left[m_{\tilde{q}}^{p}\right]\delta_{i1i2} - \frac{5}{16}\sum_p g_2^4\, \mathrm{LF}_{4,-1}\left[m_{\tilde{q}}^{p}\right]\delta_{i1i2} + \frac{2}{15}\sum_p g_2^4\, \mathrm{LF}_{5,-2}\left[m_{\tilde{q}}^{p}\right]\delta_{i1i2} -$$

$$\frac{1}{24} g_2^2 s_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1}[m_\Phi] + \frac{1}{72}\left(3 g_2^2 s_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p} + 4 g_2^4\,\delta_{i1i2}\right)\mathrm{LF}_{3,0}[m_\Phi] -$$

$$\frac{5}{48} g_2^4\, \mathrm{LF}_{4,-1}[m_\Phi]\,\delta_{i1i2} + \frac{2}{45} g_2^4\, \mathrm{LF}_{5,-2}[m_\Phi]\,\delta_{i1i2} + \frac{1}{18} g_2^4\, \mathrm{LF}_{3,0}[\tilde{\mu}]\,\delta_{i1i2} +$$

$$\frac{1}{12} g_2^4\, \mathrm{LF}_{4,-1}[\tilde{\mu}]\,\delta_{i1i2} - \frac{4}{45} g_2^4\, \mathrm{LF}_{5,-2}[\tilde{\mu}]\,\delta_{i1i2} + \frac{1}{2} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,0}\left[m_1, m_{\tilde{e}}^{p}\right] -$$

$$g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,-1}\left[m_1, m_{\tilde{e}}^{p}\right] + \frac{1}{2} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{4,1,-2}\left[m_1, m_{\tilde{e}}^{p}\right] +$$

$$\frac{1}{96} g_1^2 g_2^2\, \mathrm{LF}_{2,1,0}\left[m_1, m_{\tilde{l}}^{i1}\right]\delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2\, \mathrm{LF}_{2,2,-1}\left[m_1, m_{\tilde{l}}^{i1}\right]\delta_{i1i2} -$$

$$\frac{1}{48} g_1^2 g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_1, m_{\tilde{l}}^{i1}\right]\delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2\, \mathrm{LF}_{4,1,-2}\left[m_1, m_{\tilde{l}}^{i1}\right]\delta_{i1i2} +$$

$$\frac{1}{96} g_1^2 g_2^2\, \mathrm{LF}_{2,1,0}\left[m_1, m_{\tilde{l}}^{i2}\right]\delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2\, \mathrm{LF}_{2,2,-1}\left[m_1, m_{\tilde{l}}^{i2}\right]\delta_{i1i2} -$$

$$\frac{1}{48} g_1^2 g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_1, m_{\tilde{l}}^{i2}\right]\delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2\, \mathrm{LF}_{4,1,-2}\left[m_1, m_{\tilde{l}}^{i2}\right]\delta_{i1i2} -$$

$$\frac{1}{96} g_2^4\, \mathrm{LF}_{2,1,0}\left[m_2, m_{\tilde{l}}^{i1}\right]\delta_{i1i2} - \frac{1}{96} g_2^4\, \mathrm{LF}_{2,2,-1}\left[m_2, m_{\tilde{l}}^{i1}\right]\delta_{i1i2} - \frac{5}{48} g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_2, m_{\tilde{l}}^{i1}\right]\delta_{i1i2} +$$

$$\frac{1}{32} g_2^4\, \mathrm{LF}_{4,1,-2}\left[m_2, m_{\tilde{l}}^{i1}\right]\delta_{i1i2} - \frac{1}{96} g_2^4\, \mathrm{LF}_{2,1,0}\left[m_2, m_{\tilde{l}}^{i2}\right]\delta_{i1i2} - \frac{1}{96} g_2^4\, \mathrm{LF}_{2,2,-1}\left[m_2, m_{\tilde{l}}^{i2}\right]\delta_{i1i2} -$$

$$\frac{5}{48} g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_2, m_{\tilde{l}}^{i2}\right]\delta_{i1i2} + \frac{1}{32} g_2^4\, \mathrm{LF}_{4,1,-2}\left[m_2, m_{\tilde{l}}^{i2}\right]\delta_{i1i2} - \frac{1}{3} g_2^4\, \mathrm{LF}_{2,1,0}[m_2, \tilde{\mu}]\,\delta_{i1i2} -$$

$$\frac{1}{3} g_2^4\, \mathrm{LF}_{2,2,-1}[m_2, \tilde{\mu}]\,\delta_{i1i2} - \frac{2}{3} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4\, \mathrm{LF}_{2,2,0}[m_2, \tilde{\mu}]\,\delta_{i1i2} + \frac{1}{6} g_2^4\, \mathrm{LF}_{3,1,-1}[m_2, \tilde{\mu}]\,\delta_{i1i2} +$$

$$\frac{1}{6} g_2^4\, \mathrm{LF}_{3,2,-2}[m_2, \tilde{\mu}]\,\delta_{i1i2} + \frac{1}{3} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4\, \mathrm{LF}_{3,2,-1}[m_2, \tilde{\mu}]\,\delta_{i1i2} +$$

$$\frac{1}{8} g_2^2\left(c_\gamma\,\overline{a_e}^{pr} - s_\gamma \tilde{\mu}\,\overline{y_e}^{pr}\right)\left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu}\, y_e^{pr}\right)\mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right]\delta_{i1i2} -$$

$$\frac{1}{6} g_2^2\left(c_\gamma\,\overline{a_e}^{pr} - s_\gamma \tilde{\mu}\,\overline{y_e}^{pr}\right)\left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu}\, y_e^{pr}\right)\mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right]\delta_{i1i2} -$$

$$\frac{1}{48} g_1^2 g_2^2\, \mathrm{LF}_{2,1,0}\left[m_{\tilde{l}}^{i1}, m_1\right]\delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i1}, m_1\right]\delta_{i1i2} +$$

$$\frac{1}{48} g_2^4\, \mathrm{LF}_{2,1,0}\left[m_{\tilde{l}}^{i1}, m_2\right]\delta_{i1i2} - \frac{1}{96} g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i1}, m_2\right]\delta_{i1i2} -$$

$$\frac{1}{48} g_1^2 g_2^2\, \mathrm{LF}_{2,1,0}\left[m_{\tilde{l}}^{i2}, m_1\right]\delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i2}, m_1\right]\delta_{i1i2} +$$

$$\frac{1}{48} g_2^4\, \mathrm{LF}_{2,1,0}\left[m_{\tilde{l}}^{i2}, m_2\right]\delta_{i1i2} - \frac{1}{96} g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i2}, m_2\right]\delta_{i1i2} +$$

$$\frac{3}{8} g_2^2\left(c_\gamma\,\overline{a_d}^{pr} - s_\gamma \tilde{\mu}\,\overline{y_d}^{pr}\right)\left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu}\, y_d^{pr}\right)\mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\delta_{i1i2} -$$

$$\frac{1}{2} g_2^2\left(c_\gamma\,\overline{a_d}^{pr} - s_\gamma \tilde{\mu}\,\overline{y_d}^{pr}\right)\left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu}\, y_d^{pr}\right)\mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\delta_{i1i2} +$$

$$\frac{1}{4} g_2^2\left(s_\gamma\,\overline{a_u}^{pr} - \tilde{\mu} c_\gamma\,\overline{y_u}^{pr}\right)\left(s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma\, y_u^{pr}\right)\mathrm{LF}_{2,2,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right]\delta_{i1i2} -$$

$$\frac{1}{8} g_2^2\left(s_\gamma\,\overline{a_u}^{pr} - \tilde{\mu} c_\gamma\,\overline{y_u}^{pr}\right)\left(s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma\, y_u^{pr}\right)\mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right]\delta_{i1i2} -$$

$$\frac{1}{4} g_2^2\left(s_\gamma\,\overline{a_u}^{pr} - \tilde{\mu} c_\gamma\,\overline{y_u}^{pr}\right)\left(s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma\, y_u^{pr}\right)\mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} -$$

$$\frac{1}{4} g_2^2\left(s_\gamma\,\overline{a_u}^{pr} - \tilde{\mu} c_\gamma\,\overline{y_u}^{pr}\right)\left(s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma\, y_u^{pr}\right)\mathrm{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} +$$

$$\frac{3}{4} g_2^2\left(s_\gamma\,\overline{a_u}^{pr} - \tilde{\mu} c_\gamma\,\overline{y_u}^{pr}\right)\left(s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma\, y_u^{pr}\right)\mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} -$$

$$\frac{1}{2} g_2^2\left(s_\gamma\,\overline{a_u}^{pr} - \tilde{\mu} c_\gamma\,\overline{y_u}^{pr}\right)\left(s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma\, y_u^{pr}\right)\mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} -$$

$$\frac{5}{24} g_1^2 g_2^2\, \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_1]\,\delta_{i1i2} - \frac{1}{4} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 g_2^2\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1]\,\delta_{i1i2} +$$

$$\frac{1}{6} g_1^2 g_2^2\, \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_1]\,\delta_{i1i2} + \frac{1}{3} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 g_2^2\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1]\,\delta_{i1i2} +$$

$$\frac{2}{3} g_2^4\, \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_2]\,\delta_{i1i2} + \frac{2}{3} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4\, \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2]\,\delta_{i1i2} + \frac{1}{3} g_2^4\, \mathrm{LF}_{3,2,-2}[\tilde{\mu}, m_2]\,\delta_{i1i2} +$$

$$\frac{2}{3} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4\, \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2]\,\delta_{i1i2} - \frac{9}{8} g_2^4\, \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_2]\,\delta_{i1i2} -$$

$$\frac{7}{4} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2]\,\delta_{i1i2} + \frac{1}{2} g_2^4\, \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_2]\,\delta_{i1i2} +$$

$$m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2]\,\delta_{i1i2} - \frac{1}{8} g_2^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,-1}\left[\tilde{\mu}, m_{\tilde{e}}^{p}\right] +$$

$$\frac{1}{24} g_2^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{4,1,-2}\left[\tilde{\mu}, m_{\tilde{e}}^{p}\right] + \frac{1}{2} c_\gamma^2\,\overline{y_e}^{rp}\,\overline{y_e}^{i2s} y_e^{rs} y_e^{i1p}\, \mathrm{LF}_{2,1,0}\left[\tilde{\mu}, m_{\tilde{l}}^{r}\right] -$$

$$c_\gamma^2\,\overline{y_e}^{rp}\,\overline{y_e}^{i2s} y_e^{rs} y_e^{i1p}\, \mathrm{LF}_{3,1,-1}\left[\tilde{\mu}, m_{\tilde{l}}^{r}\right] + \frac{1}{2} c_\gamma^2\,\overline{y_e}^{rp}\,\overline{y_e}^{i2s} y_e^{rs} y_e^{i1p}\, \mathrm{LF}_{4,1,-2}\left[\tilde{\mu}, m_{\tilde{l}}^{r}\right] +$$

$$\frac{1}{8} m_1 c_\gamma g_1^2\,\overline{y_e}^{i2p}\left(-c_\gamma a_e^{i1p} + s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}\right] +$$

$$\frac{1}{16} m_1 c_\gamma g_1^2\,\overline{y_e}^{i2p}\left(-c_\gamma a_e^{i1p} + s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}\right] +$$

$$\frac{1}{8} m_1 c_\gamma g_1^2\,\overline{y_e}^{i2p}\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}\right] +$$

$$\frac{1}{8} m_1 c_\gamma g_1^2\, y_e^{i1p}\left(-c_\gamma\,\overline{a_e}^{i2p} + s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] +$$

$$\frac{1}{16} m_1 c_\gamma g_1^2\, y_e^{i1p}\left(-c_\gamma\,\overline{a_e}^{i2p} + s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] +$$

$$\frac{1}{8} m_1 c_\gamma g_1^2\, y_e^{i1p}\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] -$$

$$\frac{3}{4} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] - \frac{1}{2} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] +$$

$$\frac{1}{2} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-2}\left[m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] + \frac{1}{2} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] +$$

$$\frac{1}{16} m_1 c_\gamma g_1^2\,\overline{y_e}^{i2p}\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{e}}^{p}\right] +$$

$$\frac{1}{16} m_1 c_\gamma g_1^2\, y_e^{i1p}\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{l}}^{i2}, m_{\tilde{e}}^{p}\right] +$$

$$\frac{5}{16} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-2}\left[m_1, \tilde{\mu}, m_{\tilde{e}}^{p}\right] +$$

$$\frac{1}{16} s_\gamma \tilde{\mu} g_1^2\,\overline{y_e}^{i2p} y_e^{i1p}\,(6 m_1 c_\gamma + s_\gamma \tilde{\mu})\, \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{e}}^{p}\right] -$$

$$\frac{1}{16} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-2}\left[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}\right] - \frac{1}{16} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2\,\overline{y_e}^{i2p} y_e^{i1p}$$

$$\mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}\right] - \frac{1}{16} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-2}\left[m_1, \tilde{\mu}, m_{\tilde{l}}^{i2}\right] -$$

$$\frac{1}{16} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{l}}^{i2}\right] + \frac{1}{16} g_2^4\, \mathrm{LF}_{2,1,1,-1}\left[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right]\delta_{i1i2} -$$

$$\frac{1}{8} g_2^4 m_2^2\, \mathrm{LF}_{2,1,1,0}\left[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right]\delta_{i1i2} + \frac{1}{16} g_2^4\, \mathrm{LF}_{2,1,1,-1}\left[m_2, m_{\tilde{l}}^{i2}, \tilde{\mu}\right]\delta_{i1i2} -$$

$$\frac{1}{8} g_2^4 m_2^2\, \mathrm{LF}_{2,1,1,0}\left[m_2, m_{\tilde{l}}^{i2}, \tilde{\mu}\right]\delta_{i1i2} - \frac{1}{16} g_2^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,2,1,-2}\left[m_2, \tilde{\mu}, m_{\tilde{e}}^{p}\right] -$$

$$\frac{1}{16} s_\gamma \tilde{\mu} g_2^2\,\overline{y_e}^{i2p} y_e^{i1p}\,(2 m_2 c_\gamma + s_\gamma \tilde{\mu})\, \mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{e}}^{p}\right] +$$

$$\frac{1}{16}\left(3 g_2^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p} + g_2^4\,\delta_{i1i2}\right)\mathrm{LF}_{2,2,1,-2}\left[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}\right] +$$

$$\frac{1}{16} m_2 g_2^2\left(3 s_\gamma \tilde{\mu} c_\gamma\,\overline{y_e}^{i2p} y_e^{i1p} + g_2^2\,(m_2 + 4 s_\gamma \tilde{\mu} c_\gamma)\,\delta_{i1i2}\right)\mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}\right] +$$

$$\frac{1}{16}\left(3 g_2^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p} + g_2^4\,\delta_{i1i2}\right)\mathrm{LF}_{2,2,1,-2}\left[m_2, \tilde{\mu}, m_{\tilde{l}}^{i2}\right] +$$

$$\frac{1}{16} m_2 g_2^2\left(3 s_\gamma \tilde{\mu} c_\gamma\,\overline{y_e}^{i2p} y_e^{i1p} + g_2^2\,(m_2 + 4 s_\gamma \tilde{\mu} c_\gamma)\,\delta_{i1i2}\right)\mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{l}}^{i2}\right] +$$

$$\frac{1}{16} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{p}\right] - \frac{1}{8} g_1^2 s_\gamma^2 \tilde{\mu}^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{p}\right] +$$

$$\frac{3}{16} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i1}\right] + \frac{1}{8} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i1}\right] -$$

$$\frac{1}{8} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i1}\right] - \frac{1}{8} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i1}\right] +$$

$$\frac{3}{16} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i2}\right] + \frac{1}{8} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i2}\right] -$$

$$\frac{1}{8} g_1^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i2}\right] - \frac{1}{8} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i2}\right] -$$

$$\frac{1}{16} g_2^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{p}\right] + \frac{1}{8} g_2^2 s_\gamma^2 \tilde{\mu}^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{p}\right] -$$

$$\frac{9}{16} g_2^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i1}\right] - \frac{3}{8} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i1}\right] +$$

$$\frac{3}{8} g_2^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i1}\right] + \frac{3}{8} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i1}\right] -$$

$$\frac{9}{16} g_2^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i2}\right] - \frac{3}{8} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i2}\right] +$$

$$\frac{3}{8} g_2^2 c_\gamma^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-2}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i2}\right] + \frac{3}{8} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2\,\overline{y_e}^{i2p} y_e^{i1p}\, \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i2}\right] +$$

$$\frac{1}{16} g_1^2 g_2^2\left(-c_\gamma^2 + s_\gamma^2\right)\mathrm{LF}_{1,1,1,1,-1}\left[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right]\delta_{i1i2} +$$

$$\frac{1}{8} m_1 m_2 g_1^2 g_2^2\left(-c_\gamma^2 + s_\gamma^2\right)\mathrm{LF}_{1,1,1,1,0}\left[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right]\delta_{i1i2} +$$

$$\frac{1}{16} g_1^2 g_2^2\left(-c_\gamma^2 + s_\gamma^2\right)\mathrm{LF}_{1,1,1,1,-1}\left[m_1, m_2, m_{\tilde{l}}^{i2}, \tilde{\mu}\right]\delta_{i1i2} +$$

$$\frac{1}{8} m_1 m_2 g_1^2 g_2^2\left(-c_\gamma^2 + s_\gamma^2\right)\mathrm{LF}_{1,1,1,1,0}\left[m_1, m_2, m_{\tilde{l}}^{i2}, \tilde{\mu}\right]\delta_{i1i2} +$$

$$\frac{1}{16} g_1^2\,\overline{y_e}^{i2p}\,(m_1 c_\gamma - s_\gamma \tilde{\mu})\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] +$$

$$\frac{1}{32} g_1^2\,\overline{y_e}^{i2p}\,(m_1 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] +$$

$$\frac{1}{16} g_1^2\, y_e^{i1p}\,(m_1 c_\gamma - s_\gamma \tilde{\mu})\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] +$$

$$\frac{1}{32} g_1^2\, y_e^{i1p}\,(m_1 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] +$$

$$\frac{1}{16} g_2^2\,\overline{y_e}^{i2p}\,(m_2 c_\gamma - s_\gamma \tilde{\mu})\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] +$$

$$\frac{1}{32} g_2^2\,\overline{y_e}^{i2p}\,(m_2 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] +$$

$$\frac{1}{16} g_2^2\, y_e^{i1p}\,(m_2 c_\gamma - s_\gamma \tilde{\mu})\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] +$$

$$\frac{1}{32} g_2^2\, y_e^{i1p}\,(m_2 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] -$$

$$\frac{1}{16} g_1^2\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_1, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{i2}\right] -$$

$$\frac{1}{32} g_1^2\,\overline{y_e}^{i2p}\,(m_1 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] -$$

$$\frac{1}{32} g_1^2\, y_e^{i1p}\,(m_1 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_1, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] +$$

$$\frac{1}{16} g_2^2\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_2, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{i2}\right] -$$

$$\frac{1}{32} g_2^2\,\overline{y_e}^{i2p}\,(m_2 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] -$$

$$\frac{1}{32} g_2^2\, y_e^{i1p}\,(m_2 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_2, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] +$$

$$\frac{1}{32} g_1^2\,\overline{y_e}^{i2p}\,(m_1 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] +$$

$$\frac{1}{32} g_2^2\,\overline{y_e}^{i2p}\,(m_2 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_2, m_{\tilde{e}}^{p}, \tilde{\mu}\right] +$$

$$\frac{1}{32} g_1^2\, y_e^{i1p}\,(m_1 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i2}, m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] +$$

$$\frac{1}{32} g_2^2\, y_e^{i1p}\,(m_2 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i2}, m_2, m_{\tilde{e}}^{p}, \tilde{\mu}\right] +$$

$$\frac{1}{16} g_1^2 g_2^2\left(-c_\gamma^2 + s_\gamma^2\right)\mathrm{LF}_{2,1,1,1,-2}\left[\tilde{\mu}, m_1, m_2, m_{\tilde{l}}^{i1}\right]\delta_{i1i2} +$$

$$\frac{1}{16} m_1 m_2 g_1^2 g_2^2\left(c_\gamma^2 - s_\gamma^2\right)\mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_2, m_{\tilde{l}}^{i1}\right]\delta_{i1i2} +$$

$$\frac{1}{16} g_1^2 g_2^2\left(-c_\gamma^2 + s_\gamma^2\right)\mathrm{LF}_{2,1,1,1,-2}\left[\tilde{\mu}, m_1, m_2, m_{\tilde{l}}^{i2}\right]\delta_{i1i2} +$$

$$\frac{1}{16} m_1 m_2 g_1^2 g_2^2\left(c_\gamma^2 - s_\gamma^2\right)\mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_2, m_{\tilde{l}}^{i2}\right]\delta_{i1i2} -$$

$$\frac{1}{32} g_1^2\,\overline{y_e}^{i2p}\,(m_1 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}\right] -$$

$$\frac{1}{32} g_1^2\, y_e^{i1p}\,(m_1 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] -$$

$$\frac{1}{32} g_2^2\,\overline{y_e}^{i2p}\,(m_2 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma a_e^{i1p} - s_\gamma \tilde{\mu}\, y_e^{i1p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}\right] -$$

$$\left.\frac{1}{32} g_2^2\, y_e^{i1p}\,(m_2 c_\gamma + s_\gamma \tilde{\mu})\left(c_\gamma\,\overline{a_e}^{i2p} - s_\gamma \tilde{\mu}\,\overline{y_e}^{i2p}\right)\mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right]\right)$$